

Examples of two elementary functions in Python

► [Loading the compressed file of codes and data \[folder:"ADT_py"\]](#)

The codes of these two elementary functions and their contexts of use are pedagogical examples of the book *Analyse des Données textuelles* [Analysis of Textual Data] (L. Lebart, B. Pincemin, C. Poudat), Presses de l'Université du Québec [in French], 2019, in press.

The Python programming language, whose first version dates back to Guido van Rossum in 1989, is the versatile tool expected by researchers who are working on texts. While being easy to access, this open source language provides a kind of synthesis between script languages such as Perl and classical object-oriented languages such as C++ or Java. The user can download Python (and for more comfort its IDLE interface) from <https://www.python.org> and learn interactively using the help buttons available.

The following programs obviously assume that the reader has some knowledge of the basic notions of language. These can be acquired interactively by consulting the Help downloaded with Python (buttons: Help, then Tutorial of the IDLE interface).

The following basic reference cards could be useful (as a terse summary) for beginners:

Example 1 of Python memento card:

<https://perso.limsi.fr/poinal/python:memento>

Example 2 of Python memento card:

<https://www.cs.put.poznan.pl/csobaniec/software/python/py-qrc.html>

A wealth of books / manuals for learning Python are available on the web.

1- Computation and printing of a lexical table (words X texts)

2- Computation and printing of a concordance from a series of texts



1- Computation and printing of a lexical table (words X texts) from a series of texts.

This is simply to penetrate, thanks to the Python language, in the black box of a commonplace functionality available in most statistical software of text analysis.

Evidently, the code could be much more compact, but perhaps less readable.

Commands for the Python interpreter (IDLE for example)

```
#-----
import os                # os module
os.chdir("c:/ADT_py")    # name of the folder containing
                        # program and data, in the root "c:/" in this example
chemin = "poem.txt"      # name of text file (same folder)
import table_lex_E       # program file: table_lex_E.py
                        # (same folder)
from table_lex_E import * # import functions
#-----
tablex(chemin, 2)        # executing function tablex included
                        # in the downloaded file: table_lex_E.py
# (with: seuil = 2: default value for the frequency threshold).
#-----
```

The small data-test file “poem.txt” is provided in the folder ADT_py.

[Texts in DtmVic format: separators of texts: “*****” (beginning of line) followed by 4 blank spaces and text titles. End of text: “====”(beginning of line)].

```
*****    LAMARTINE
voilà les feuilles sans sève,
qui tombent sur le gazon
voilà le vent qui s'élève,
et gémit dans le vallon
voilà l'errante hirondelle,
qui rase du bout de l'aile,
l'eau dormante des marais...
voilà l'enfant des chaumières,
qui glane sur la bruyère,
le bois tombé des forêts...
*****    GAUTIER
l'automne va finir, au milieu du ciel terne,
dans un cercle blafard et livide que cerne
un nuage plombe, le soleil dort. du fond
des étangs remplis d'eau monte un brouillard qui fond
collines, champs, hameaux dans une même teinte.
sur les carreaux la pluie en larges gouttes tinte.
*****    VERLAINE
les sanglots longs
des violons
de l'automne
blessent mon coeur
d'une langueur
```

monotone.
 tout suffocant
 et blême, quand
 sonne l'heure,
 je me souviens
 des jours anciens
 et je pleure.

**** BRUGNOT

l'herbe se fane dans les près,
 les jours de soleil sont passés
 les feuilles jaunes et pourprées
 jonchent les sentiers effacés.
 le voilà donc mon bel automne.

**** BAUDELAIRE

bientôt nous plongerons dans les froides ténèbres
 adieu, vive clarté de nos étés trop courts.
 j'entends déjà tomber avec des chocs funèbres
 le bois retentissant sur le pavé des cours.

=====

This mini-corpus is only intended to verify the proper functioning of the code. The same code will provide the lexical table of the STATE OF THE UNION corpus (corpus about 2000 times larger) in seconds (in the latter case, it is prudent to start with a minimum frequency threshold of 200 for words). The table below is the image of the "tablexfile.txt" file produced by the function tablex().

Lexical table crosstabulating Words and Poems

	LAMARTI	GAUTIER	VERLAIN	BRUGNOT	BAUDELA
automne	0	1	1	1	0
bois	1	0	0	0	1
d	0	1	1	0	0
dans	1	2	0	1	1
de	1	0	1	1	1
des	3	1	2	0	2
du	1	2	0	0	0
eau	1	1	0	0	0
et	1	1	2	1	0
feuilles	1	0	0	1	0
fond	0	2	0	0	0
je	0	0	2	0	0
jours	0	0	1	1	0
l	4	1	2	1	0
la	1	1	0	0	0
le	4	1	0	1	2
les	1	1	1	4	1
mon	0	0	1	1	0
qui	4	1	0	0	0
soleil	0	1	0	1	0
sur	2	1	0	0	1
un	0	3	0	0	0
une	0	1	1	0	0
voilà	4	0	0	1	0



2- Computation and printing of a concordance from a series of texts.

Commands for the Python interpreter (IDLE for example)

```
#-----
import os                                # os module
os.chdir("c:/ADT_py")                   # designates the folder containing
program and data/
chemin = " SOTU_40_08.txt "             # name of text file (same folder)
# (discourses: STATE OF THE UNION from 1940 to 2008)
import concord_E                         # program file: concord_E.py (same
folder)
from concord_E import *                  # call of functions in concord_E.py
#-----
cible = "dream"                         # "cible" means "target word"
conco (chemin, cible)                   # executing function  conco
included in the file: concord_E.py
#-----
```

The table below is the image of the "concordance.txt" file produced by the function conco().
Output table (excerpt: 4 last presidents up to 2008) of results for pivot « dream » in the corpus STATE OF THE UNION.

Concordance table [KWIC] for the word "dream"

```
**** -----      41BUSH
17825                families achieve the      dream of home
ownership. but make no mistake,
17898 your living rooms, hold fast to your      dreams because
ultimately America's future r
17942 t century. our nation is the enduring      dream of every
immigrant
17961 he future we can make for ourselves. but    dreams alone won't
18095 uture, every kid is the same: full of      dreams, ready to
take on the world, all spe
18104 on on a new century, your century, on      dreams we cannot
see, on the destiny
18324                                have the bad    dreams children
once had in decades past. t
18408      real estate. for those Americans who    dream of buying a
first
**** -----      42CLINTON
18569 enter it having secured the American      dream for ourselves
and for future
18620                                billion to make the  dream of
enterprise zones real, we propose to
```

18897	ation, and a fair shot at the American	dream, they will
	do extraordinary things. we	
18904	working, the American	dream has been
	slipping away. in 1992 the amer	
19286	on a mission: to restore the American	dream for all our
	people and to make sure th	
19494	n entrepreneurs are living the American	dream. if we want
	it to stay that way, those	
19714	promise of this country, the enduring	dream from that
	first and most-sacred cove	
19734	ions: first, how do we make the American	dream of
	opportunity for all a reality for all	
19768	our individual	dreams must be
	realized by our common effo	
19961	resources, and even	dreams. bosnia and
	we stood up for peace in bo	
20204	ll cultures. this will no longer be a	dream, but a
	necessity. and over	
20672	me here to work for their own American	dreams. let's keep
	our cities going strong in	
21129	union of our founders'	dreams. we are
	now, at the end of a centu	
21144	he more perfect union of our founders'	dreams.
21167	nt, America again has the confidence to	dream big dreams.
	but we must not let our	
21168	complacency. we will be judged by the	dreams and deeds
	we pass on to our children	
21231	their test scores. to make the American	dream achievable
	for all, we must make colle	
21532	we remain a new nation. as long as our	dreams outweigh
	our memories,	
****	----- 43BUSH	
21683	d in the way of families achieving their	dreams. the surplus
	is not the	
22540	not punish, the efforts and	dreams of
	entrepreneurs. small business is	
22769	we live in the country where the biggest	dreams are born.
22770	the abolition of slavery was only a	dream until it
	was fulfilled. the liberatio	
22771	fall of imperial communism was only a	dream until, one
	day, it was accomplished.	
22772	our generation has	dreams of its own,
	and we also go forward w	
22776	founding ideals and carried on a noble	dream. tonight we
	are comforted by the hope o	
22797	dom's cause. far from being a hopeless	dream, the advance
	of freedom	
23027	raise their sights and achieve their	dreams. a hopeful
	society comes to the aid of	
23196	blood and bodies to put an end to your	dreams, and what
	is coming is even worse. a	
23397	ountry, there are boys and girls with	dreams and a decent
	education	
****	----- 44OBAMA	
23645	you built your	dreams upon that's
	now hanging by a thread; t	
23902	he most power or celebrity, but from the	dreams and
	aspirations of	
24004	when an entrepreneur takes a chance on a	dream, or a worker
	decides its time she became	

24273 ertain; to do what it took to keep the dream of this
nation alive for their
24286 this moment to start anew, to carry the dream forward,

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